

Climate Resilience: A Multi-Source Analysis of Climate Policy Evolution since the Paris Agreement (2013–2025)

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This application report documents a data mining pipeline bringing together three separate sources of climate data: the Climate Policy Database (CPDB), the UNFCCC NDCs and the Notre Dame Global Adaptation Initiative (ND-GAIN). The exploration analyzes the evolution of climate strategies since the Paris agreement (2015), with a particular focus on the integration of adaptation measures alongside mitigation efforts.

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1 Introduction

Climate policy conversations are historically focused on the mitigation of climate change. However, the increasing frequency and severity of climate impacts has increased the importance of adaptation measures in national policy portfolios. The Paris Agreement (2015) marked a pivotal moment in global climate governance, with its dual focus on mitigation and adaptation.

International climate databases systematically under represent adaptation policies, particularly in the Global South where adaptation is a *survival imperative* [1, p.10]. When adaptation is invisible in the data, it becomes invisible in policy debates, climate finance allocation and accountability mechanisms. The lack of “comparative global analysis and systematic scrutiny of national adaptation legislation and policies has hampered the ability to assess progress, identify gaps, and understand effective frameworks” [1, p.10]. This project serves as a digital audit of climate ambition across our selection of countries with an aim to discover if and how countries are shifting towards adoption adaptation into the mix.

The Global North have historically been responsible for the majority of emissions and have more resources to invest in both mitigation and adaptation. The Global South, whilst being less responsible for the acceleration of climate change, are the ones who bear the brunt of it. The comparative analysis of the Global North and Global South is therefore critical to understanding the equity implications of climate policy evolution.

2 Research Question

Has there been a shift towards climate adaptation adoption in policy portfolios after the Paris Agreement, and does this shift differ between Global North and Global South countries?

Key Terms:

- **Climate Policy Portfolio:** The collection of all climate-related policy strategies enacted by a country.
- **Mitigation:** Efforts to reduce or prevent the emission of greenhouse gases to limit global warming.
- **Adaptation:** Efforts to adjust to actual or expected climate change and its effects.

Boundaries:

Temporal Boundary (2013–2025)

The temporal coverage allows us to capture the policy landscape leading up to the Paris Agreement, providing a baseline for comparison. The end point of 2025 ensures we have the most recent data available.

Country Selection

The selection of 12 countries is designed to provide a diverse sample that includes both Global North and Global South nations.

The logic is as follows:

Country	ISO-3 Code	Region	ND-GAIN Resilience
Canada	CAN	North	High
Germany	DEU	North	High
Japan	JPN	North	High
United States	USA	North	High
United Kingdom	GBR	North	High
Denmark	DNK	North	High
Saudi Arabia	SAU	South	Medium
Chile	CHL	South	Medium
India	IND	South	Low
South Africa	ZAF	South	Low
Colombia	COL	South	Low
Rwanda	RWA	South	Low

Note: The USA has left the Paris agreement more than once, I have included it simply out of interest

3 Data Sources

3.1 NewClimate Institute’s Climate Policy Database (CPDB) [2]

The CPDB provides the baseline for actual policy adoption. It offers broad geographic coverage. Its primary limitation is a “Global North bias,” where data coverage is stronger. This is a gap addressed by the inclusion of NDC texts. The CPDB’s provides structured policy data with policy types (mitigation, adaptation), decision dates and country codes, allowing for a quantitative analysis of policy trends over time.

The cleaning and processing of the CPDB data is explained in issue #7.

3.2 UNFCCC Nationally Determined Contributions (NDCs) [3]

These documents serve as primary statements of government priority. While they allow for a nuanced NLP analysis of “Lexical Hardness” (Hard vs. Soft law) as well as “Adaptation focus” (Percentage of text focused on adaptation), they represent diplomatic pledges rather than tangible enforcement. The plus side of using NDCs is that each country is able to represent themselves as they choose, removing the reliance on third-party data collection and coding which may be biased or incomplete, particularly for the Global South.

The collection and processing of the NDCs is explained in issue #8.

3.3 ND-GAIN Index (Resilience Benchmarking) [4]

The ND-GAIN Index provides the context of necessity. The ND-GAIN scores are a composition of vulnerability and readiness indicators. While the Vulnerability score measures physical exposure, the overall Index balances this against a country’s Readiness (its economic, social, and governmental capacity to deploy adaptation solutions). The overall index is therefor a measure of resilience.

The tidying of nd-gain is explained in issue #12.

4 Technological Approach & Challenges

The entire workflow is orchestrated by a single master script (`main.R`) that manages:

1. **Dependency installation**
2. **Data acquisition**
3. **Data processing** (cleaning, merging, feature engineering).
4. **Analysis**

5. **Output generation** (figures, tables, this report).
6. **Environment Cleaning** (removing temporary files and data).

Data Acquisition

To implement a Multi-Source Triangulation strategy I first bridged R with Python (reticulate) to access the CPDB API, then developed a custom Scraping Pipeline for UNFCCC NDCs. This involved automated PDF downloading and text extraction via pdftools, allowing for the inclusion of primary-source data from countries often under-reported in secondary databases (e.g., Rwanda, Colombia).

Key challenge:

I had initially planned to focus my analysis on the OECD CAPMF database which provides stringency scores for climate policies. This is an extremely detailed and useful source of data which required extensive cleaning in order to create a structured data frame. However I subsequently realized that the CAPMF data only covers a subset of the countries in my sample and is also mainly a database for mitigation policies, so I made the decision to shift my focus towards the CPDB which has a broader coverage of both countries and policy types. *Discussed in Issue #2*

Lexical Audit & Feature Engineering

NDC Strategic intent was quantified through a Keyword-Based Heuristic. I developed two distinct lexicons:

- Focus Score: A ratio of adaptation-to-mitigation terminology normalized by document length.
- Ambition Score: A “Lexical Hardness” index comparing mandatory (“shall”, “must”) vs. aspirational (“aim”, “should”) language, utilizing a smoothing constant (+0.001) to handle zero-count denominators in shorter documents.

These are discussed in issue #15, #5 and #4.

5 Results

5.1 Global Adaptation Growth

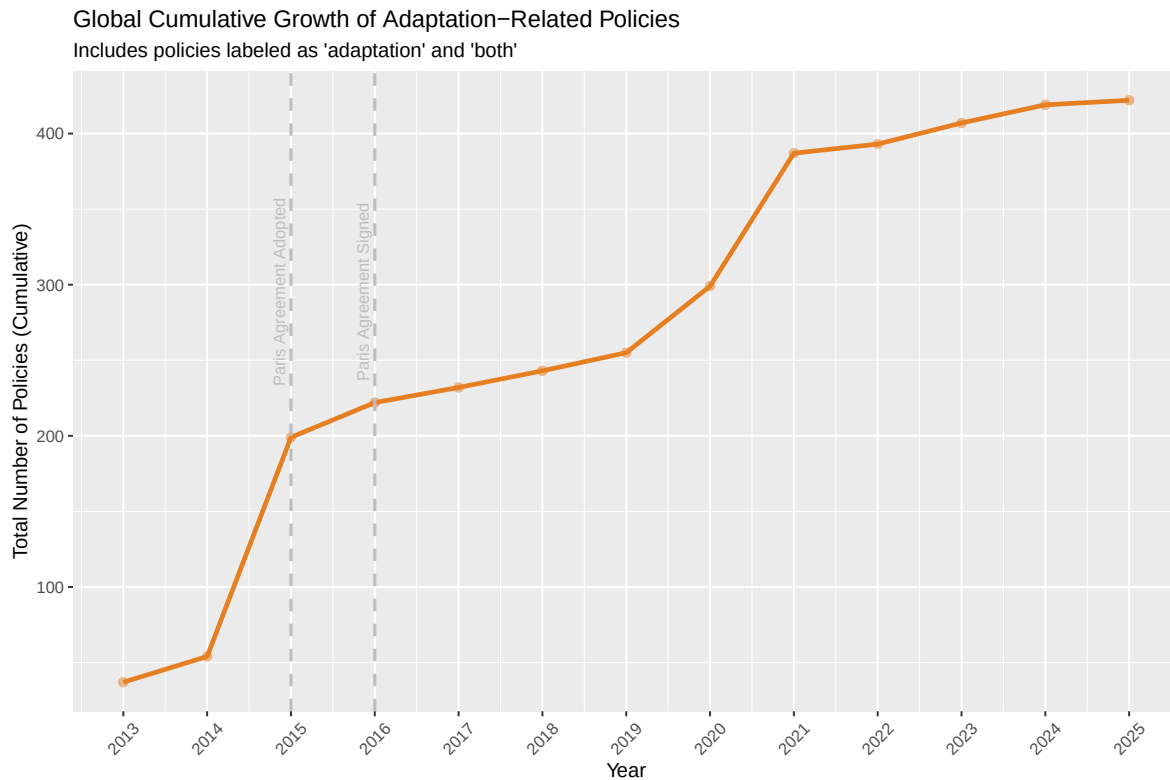


Figure 1: Global Cumulative Growth of Adaptation Policies (2013-2025)

Initial analysis of climate policies in the cpdb reveals a visible acceleration in global adaptation policy adoption following the 2015 Paris Agreement. This is the starting point for our analysis, confirming that the Paris Agreement did indeed catalyze a shift in policy focus. This already answers the first part of our research question (*Has there been a shift towards climate adaptation adoption in policy portfolios after the Paris Agreement?*).

5.2 Policy Implementation per Country (The Adaptation Mix)

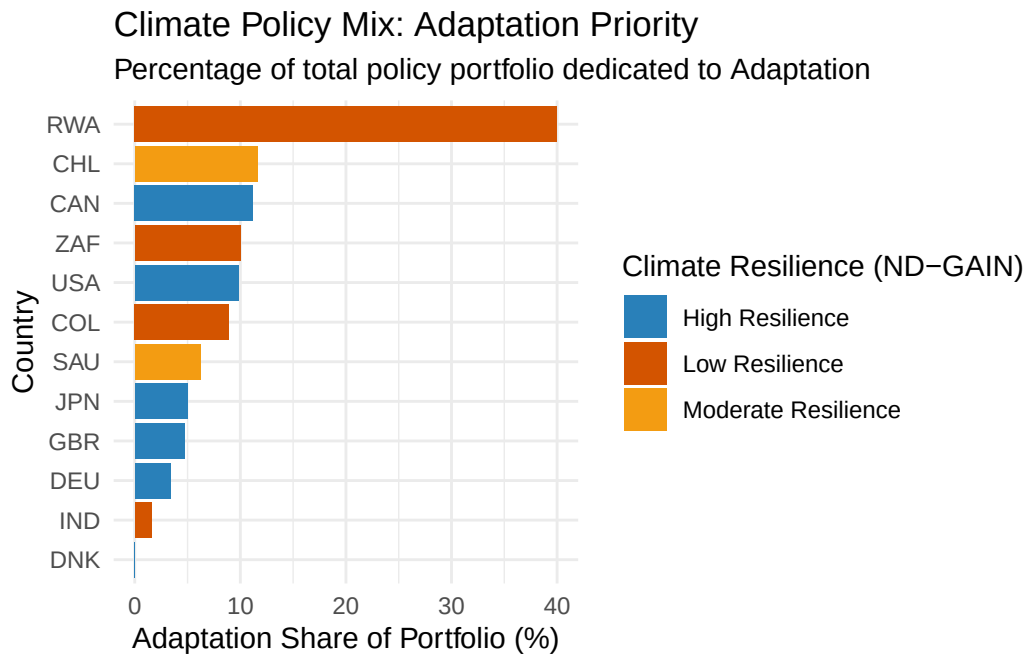


Figure 2: The Adaptation Mix: Percentage of Portfolio focusing on Adaptation

Zooming in to our chosen list of countries (Figure 2) we are able to see the spread of each policy portfolio. What percentage of implemented policies focus on adaptation. The ND-GAIN scores have been added in order to visualize the comparison of vulnerability. While some vulnerable nations like Rwanda show a high adaptation ratio, others like India lag behind. This would suggest that vulnerability may not be a driving factor in adaptation adoption.

However, it is important to keep in mind that this data set has its limitations. Two problems 1) the tracking of adaptation is historically complex and most often under reported, particularly for the Global South and 2) the CPDB data has a bias towards Global North countries. *Issue #1 discusses the choice to move away from absolute policy counts towards a relative ratio in order to get a clearer picture.*

5.3 Lexical Adaptation Focus in NDCs (The Responsibility Gap)

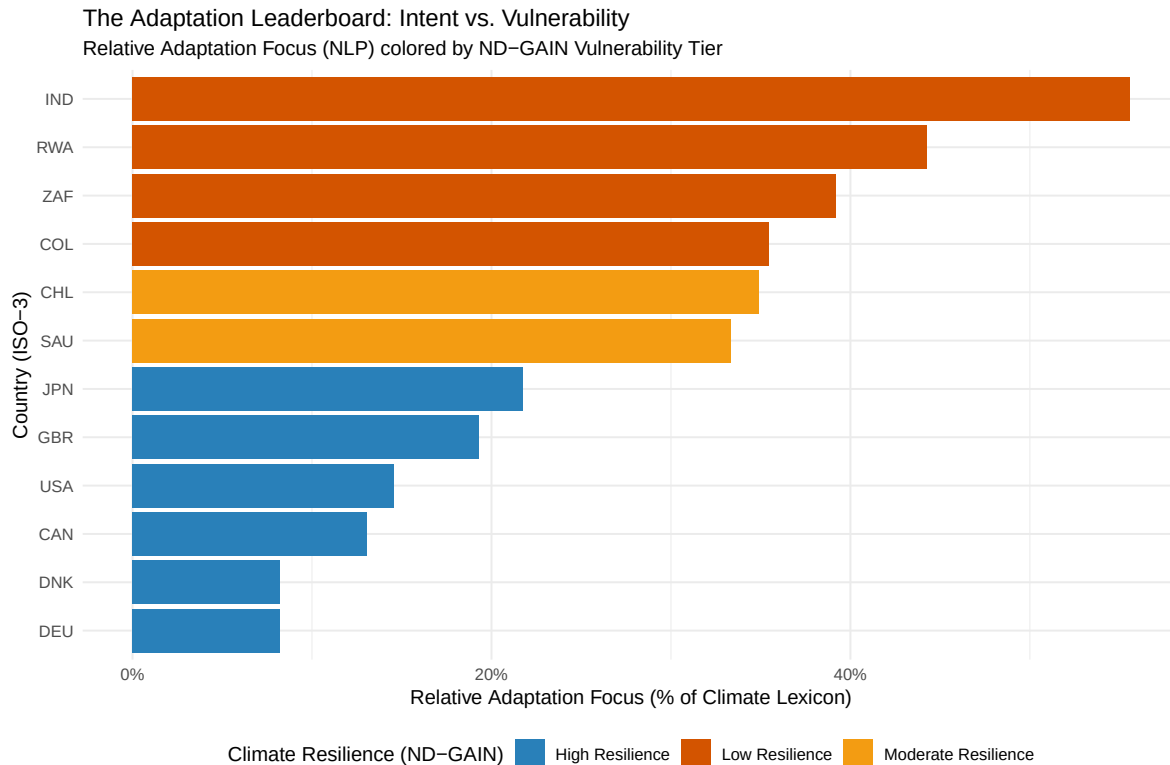


Figure 3: Adaptation Focus in NDCs

By shifting to an NLP-based analysis of each Nationally Determined Contribution (NDC) (Figure 4), the correlation between physical risk and strategic intent becomes obvious. High-vulnerability countries dedicate significantly more “mindshare” to adaptation in their NDCs compared to the Global North. This is the Responsibility Gap in action: the nations least responsible for historical emissions are the ones forced to prioritize survival strategies.

Comparing Policy Implementation (CPDB) to Strategic Focus (NDC)			
ISO-3	CPDB Mix (%)	NDC Focus (%)	The Gap (diff)
IND	1.6	55.6	54.0
RWA	40.0	44.3	4.3
ZAF	10.0	39.2	29.2
COL	8.9	35.4	26.5
CHL	11.6	34.9	23.3
SAU	6.2	33.3	27.1
JPN	5.0	21.7	16.7
GBR	4.7	19.3	14.6
USA	9.8	14.6	4.8
CAN	11.2	13.0	1.8
DEU	3.4	8.2	4.8
DNK	0.0	8.2	8.2

Figure 4: Table 1: The Ambition Audit

Table 1 shows a comparison between the CPDB policy adaptation mix and the NDC lexical adaptation focus. The gap between intent and policy implementation is particularly wide for the Global South, where adaptation is a critical need but resource constraints and governance (readiness) may limit actual policy adoption. This is the Commitment Gap: vulnerable countries are prioritizing adaptation in their strategic documents but are unable to translate this into binding policies. A further understanding of this commitment gap is provided by figure 6, which maps the relative adaptation focus against the lexical hardness of the NDC language.

5.4 Adaptation Focus vs. Ambition/Stringency in NDCs (The Commitment Gap)

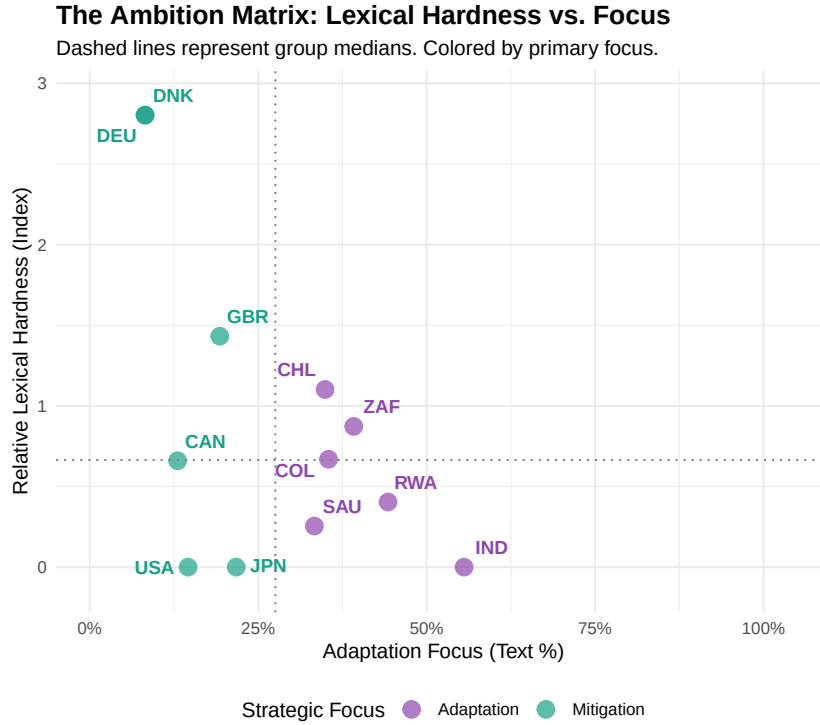


Figure 5: Climate Ambition Matrix: Lexical Hardness vs. Adaptation Focus

The Ambition Score reflects the relative frequency of mandatory language (e.g., shall, must, mandate) versus aspirational terms (e.g., should, aim, aspire), normalized against the group mean. While lexical hardness varies across all regions, the strategic focus remains deeply polarized. By centering the matrix on group medians, the split is visually stark: the Global South and the Global North are legislating for fundamentally different realities.

6 Discussion

To answer the research question (*Has there been a shift towards climate adaptation adoption in policy portfolios after the Paris Agreement, and does this shift differ between Global North and Global South countries?*) there has been a shift towards climate adaptation adoption in policy portfolios after the Paris Agreement and this shift differs significantly between Global North and Global South countries. The Global South, which is more vulnerable to climate impacts, shows a higher focus on adaptation in their NDCs but faces significant challenges in translating this into binding policies, as evidenced by the Commitment Gap. The Global North, while having more resources, shows less focus on adaptation both in their NDCs and in actual policy implementation.

The Responsibility and Commitment Gaps both provide important evidence that it is those who are most vulnerable to climate impacts (the Global South) who are spending more of their “mind share” on adaptation. This illustrates the injustice at the heart of climate change: those who are least responsible for causing it are the ones who suffer its worst effects and have the greatest need for adaptation, yet they have the least capacity to do so.

The Commitment Gap finding has particular significance. It demonstrates that Global South adaptation priorities are substantive (evidenced by high keyword frequency and development plans explicitly linking adaptation to poverty reduction [1, p.14-15]), but these priorities are constrained by resource limitations (manifested as soft law language because commitments are conditional on international finance [1, p.14]).

7 Limitations

The fundamental challenge of tracking adaptation in policy is perhaps this project's greatest limitation. However, every effort has been made to account for this.

A primary limitation is the reliance on the CPDB for policy data, which is not exhaustive and has a bias toward Global North countries. This means that the adaptation policies of many Global South countries may be underrepresented, potentially skewing the analysis. The use of relative ratios instead of exact counts is an effort to avoid this bias. The temporal lag of policy data is another concern, as there may be a delay between the strategic intent expressed in NDCs and the actual implementation of policies.

The NDC analysis is limited due to the variability in language and the challenge of quantifying qualitative commitments. The execution of the Lexical Audit was constrained by the depth of the text-mining heuristics. The current Keyword-Based Heuristic assumes that word frequency correlates with political priority. While normalized for document length, this does not account for the context of mentions *Discussed in issue #5 and issue #4*. Future iterations would benefit from a more robust NLP framework to move beyond simple frequency counts and more accurately distinguish between aspirational rhetoric and enforceable domestic mandates.

8 References

- [1] Grantham Research Institute on Climate Change and the Environment. (2025). *Climate change adaptation laws and policies: Global trends and insights 2025*. London School of Economics and Political Science. <https://www.lse.ac.uk/granthaminstitute/wp-content/uploads/2026/02/Climate-change-adaptation-laws-and-policies.pdf>
- [2] NewClimate Institute, Wageningen University and Research, & PBL Netherlands Environmental Assessment Agency. (2024). *Climate Policy Database* [Data set]. Zenodo. <https://doi.org/10.5281/zenodo.7774109>
- [3] United Nations Framework Convention on Climate Change. (2026). *NDC Registry* [Database]. <https://unfccc.int/NDCREG>
- [4] University of Notre Dame. (2026). *Notre Dame Global Adaptation Initiative Country Index* [Data set]. <https://gain.nd.edu/our-work/country-index/>

9 LLM Declaration

The following LLMs have been used in the creating of this report and project:

- Gemini for technical troubleshooting, code review and idea exploration: I used Gemini throughout the coding process and also as a way to talk through my ideas. All code has been a joint effort, never a complete copy-paste but more of a back and forth discussion about best methods. I have a background in software development but have never used R before, so Gemini has helped me navigate the syntax.
- Github Copilot for code suggestions and auto completion: I used Copilot to speed up coding and get suggestions. Again, all code has been a joint effort, with me writing the logic and Copilot suggesting syntax and functions.
- Scispace for help writing this report: I have used Scispace to help organize my thoughts and ensure that I am covering all the necessary sections and points.

Total Word Count (minus LLM Declaration): 2202